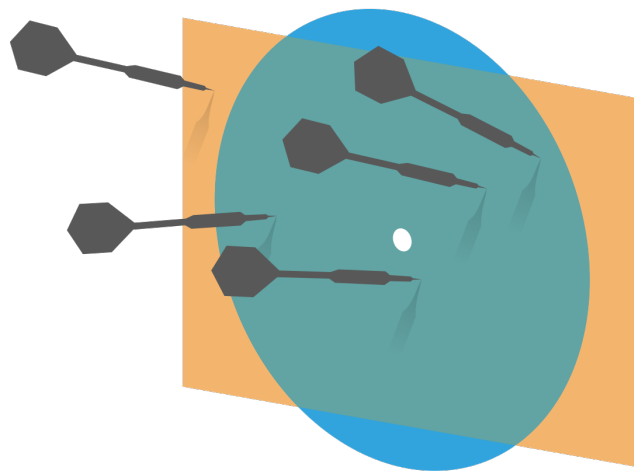


Probability and Statistics for Computer Science



Can we call e the
exciting e ?

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n$$

Credit: wikipedia

Last time

- ✱ Expected value of continuous distribution

$$E[X] = \int_{-\infty}^{\infty} x p(x) dx$$

$$p(x) dx = P(x < X < x+dx)$$

- ✱ CDF of continuous distribution

$$P(X < x) = \int_{-\infty}^x p(x) dx$$

$$\int_{-\infty}^{\infty} p(x) dx = 1$$

- ✱ Continuous uniform distribution

$$P(a < X < b) = \int_a^b p(x) dx$$

- ✱ Poisson distribution

$$P(X=k) = \frac{e^{-\lambda} \lambda^k}{k!} \quad k=0,1,\dots$$

- ✱ Exponential distribution

$$p(x) = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Objectives

- ✱ Independence of conti. RVs
- ✱ Normal (Gaussian) distribution

Independence of two continuous RVs

$$p(x, y) \cdot dx dy = p(x) \cdot dx \cdot p(y) dy$$

$$\Rightarrow p(x, y) = \underline{p(x)} \underline{p(y)}$$

↳ joint density

↘ marginal
density

Normal (Gaussian) distribution

- ✱ The most famous continuous random variable distribution. The probability density is this:

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \quad x \in (-\infty, \infty)$$

↪ e



Carl F. Gauss
(1777-1855)
Credit: wikipedia

$$\int_{-\infty}^{\infty} p(x) dx = 1$$

DeGroot
pg 304

Normal (Gaussian) distribution

- ✱ The most famous continuous random variable distribution. The probability density is this:

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$



Carl F. Gauss
(1777-1855)
Credit: wikipedia

$$E[X] = \mu \quad \& \quad \text{var}[X] = \sigma^2$$

$$\begin{aligned} & \int f(x) g'(x) dx \\ &= f(x) g(x) - \int f'(x) g(x) dx \\ E[X] &= \int_{-\infty}^{\infty} x p(x) dx \\ &= \mu \end{aligned}$$

Normal (Gaussian) distribution

- ✱ The most famous continuous random variable distribution.

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

$$\int_{-\infty}^{+\infty} p(x) dx = 1$$

$$E[X] = \mu \quad \& \quad \text{var}[X] = \sigma^2$$

$$\text{var}[x] = \int_{-\infty}^{\infty} x^2 p(x) dx - (E[x])^2$$



Carl F. Gauss
(1777-1855)
Credit: wikipedia

Normal (Gaussian) distribution

- ✱ A lot of data in nature are approximately normally distributed, ie. **Adult height**, etc.

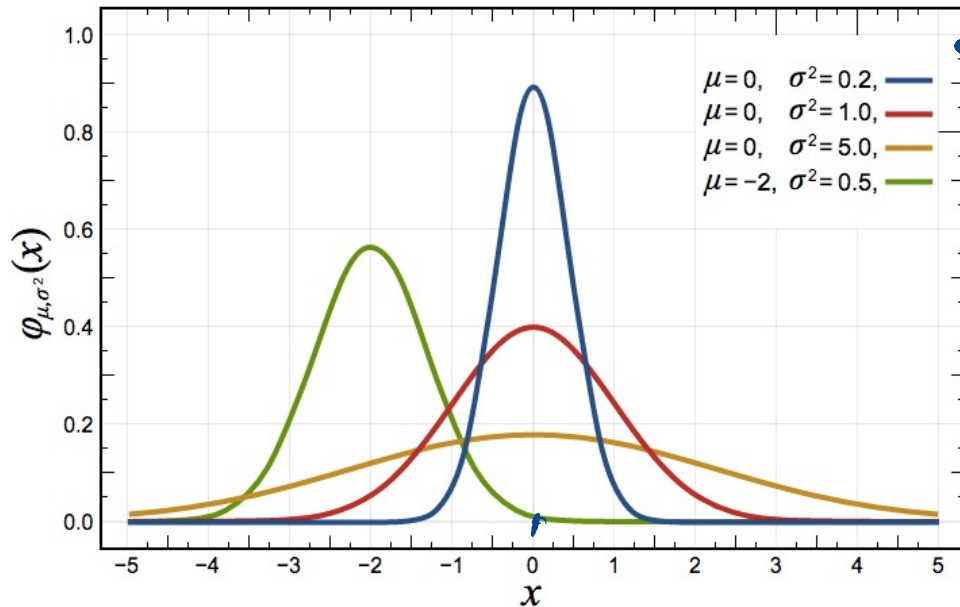
$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$



Carl F. Gauss
(1777-1855)
Credit: wikipedia

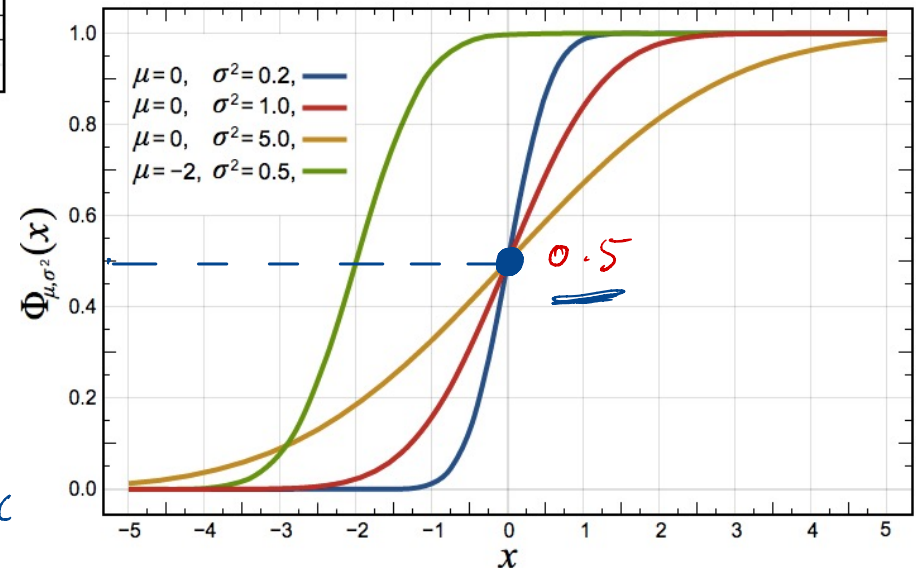
$$E[X] = \mu \quad \& \quad var[X] = \sigma^2$$

PDF and CDF of normal distribution curves



← PDF $p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

$F(x) = \int_{-\infty}^x p(x) dx$
 $\parallel$
 $P(X \leq x)$ CDF

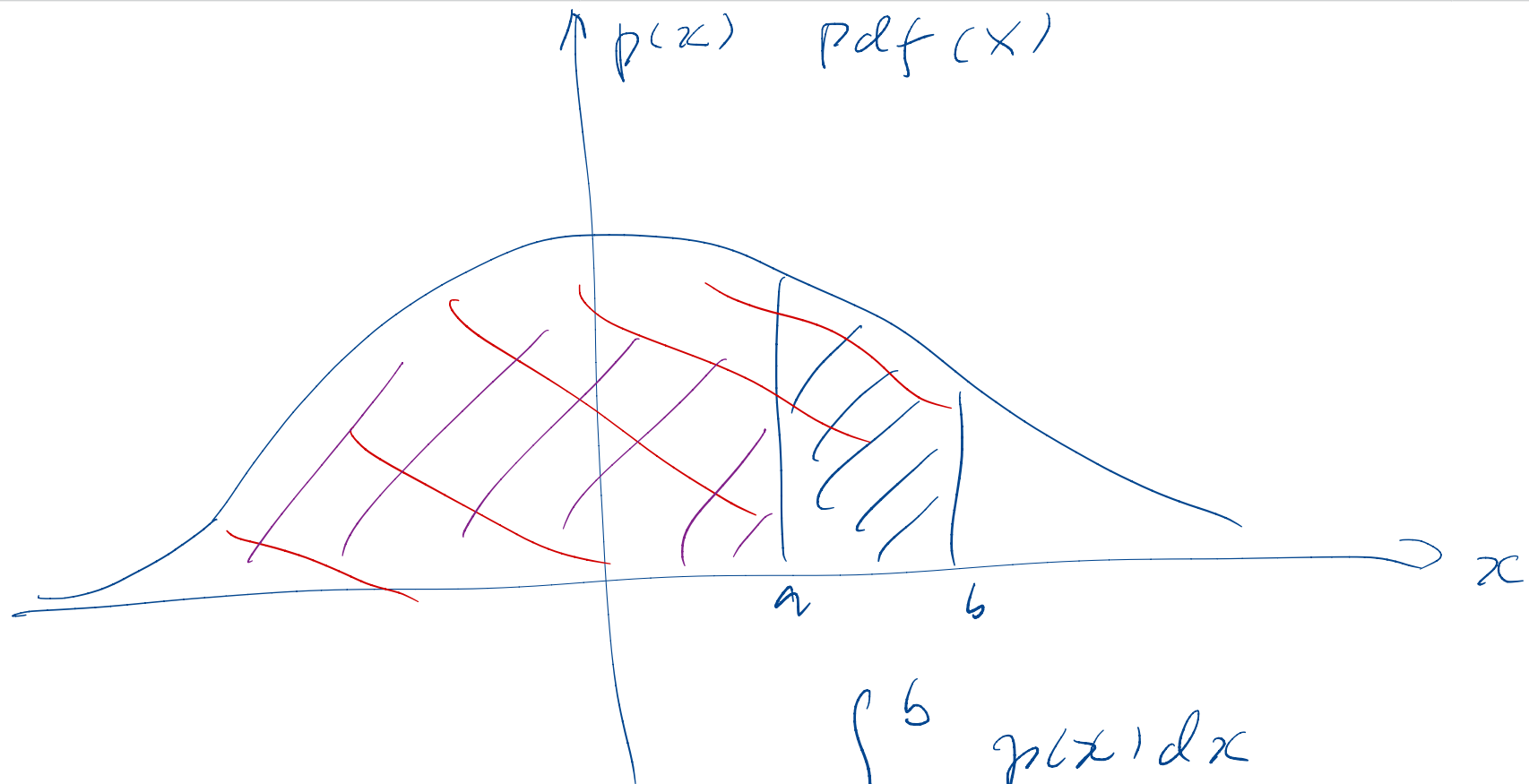


$p(a < x < b) = ?$

$\int_a^b p(x) dx$

$= \int_{-\infty}^b p(x) dx - \int_{-\infty}^a p(x) dx$

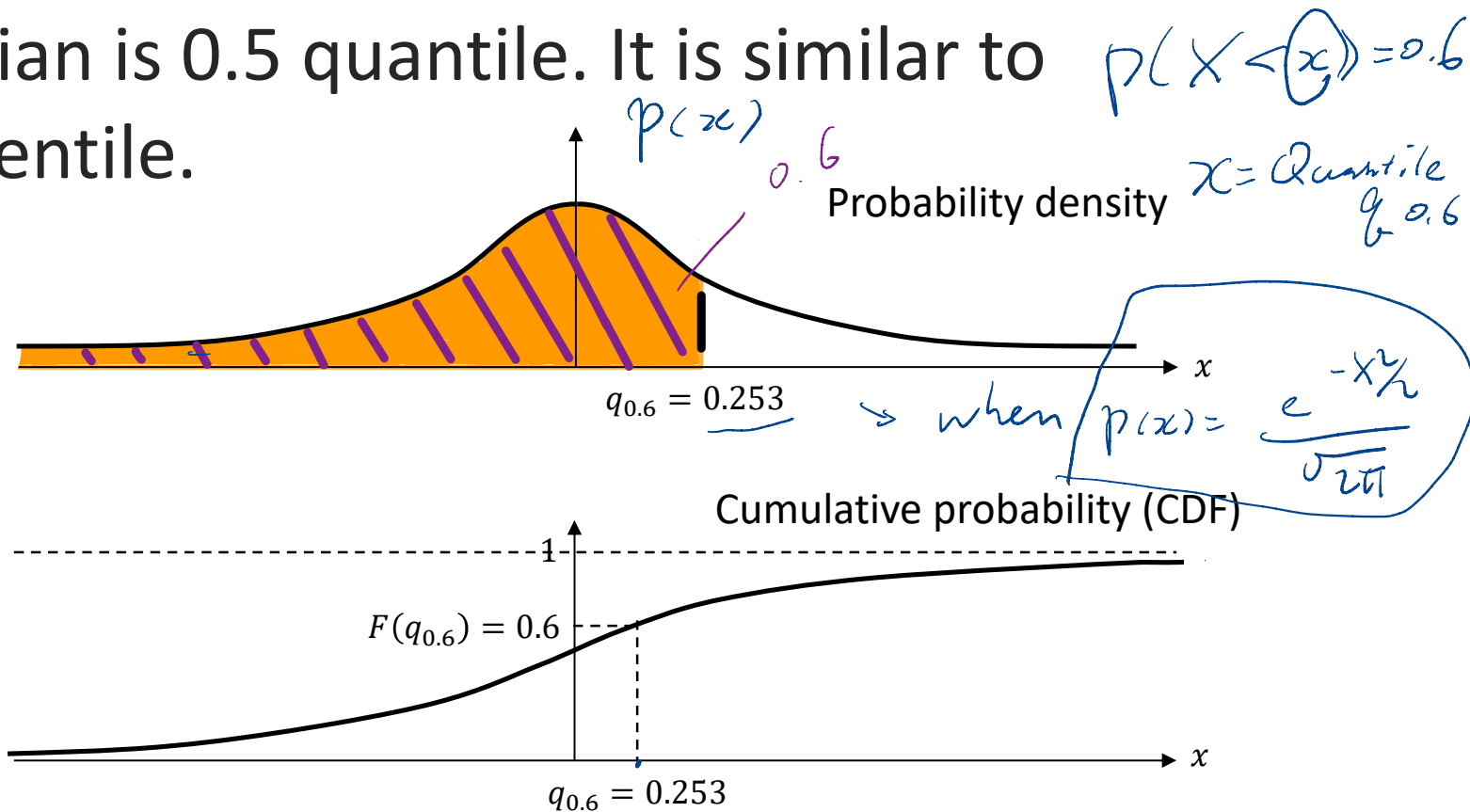
Credit: wikipedia



$$\begin{aligned} & \int_a^b p(x) dx \\ &= \int_{-\infty}^b p(x) dx - \int_{-\infty}^a p(x) dx \\ &= F(b) - F(a) \end{aligned}$$

Quantile

- ✱ Quantiles give a measure of location, the median is 0.5 quantile. It is similar to percentile.



Q.

✱ What is the value of 50% quantile in a standard normal distribution?

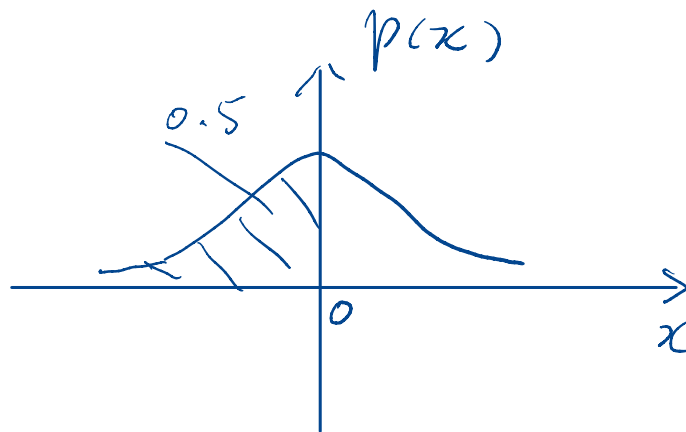
A. -1

☒ B. 0

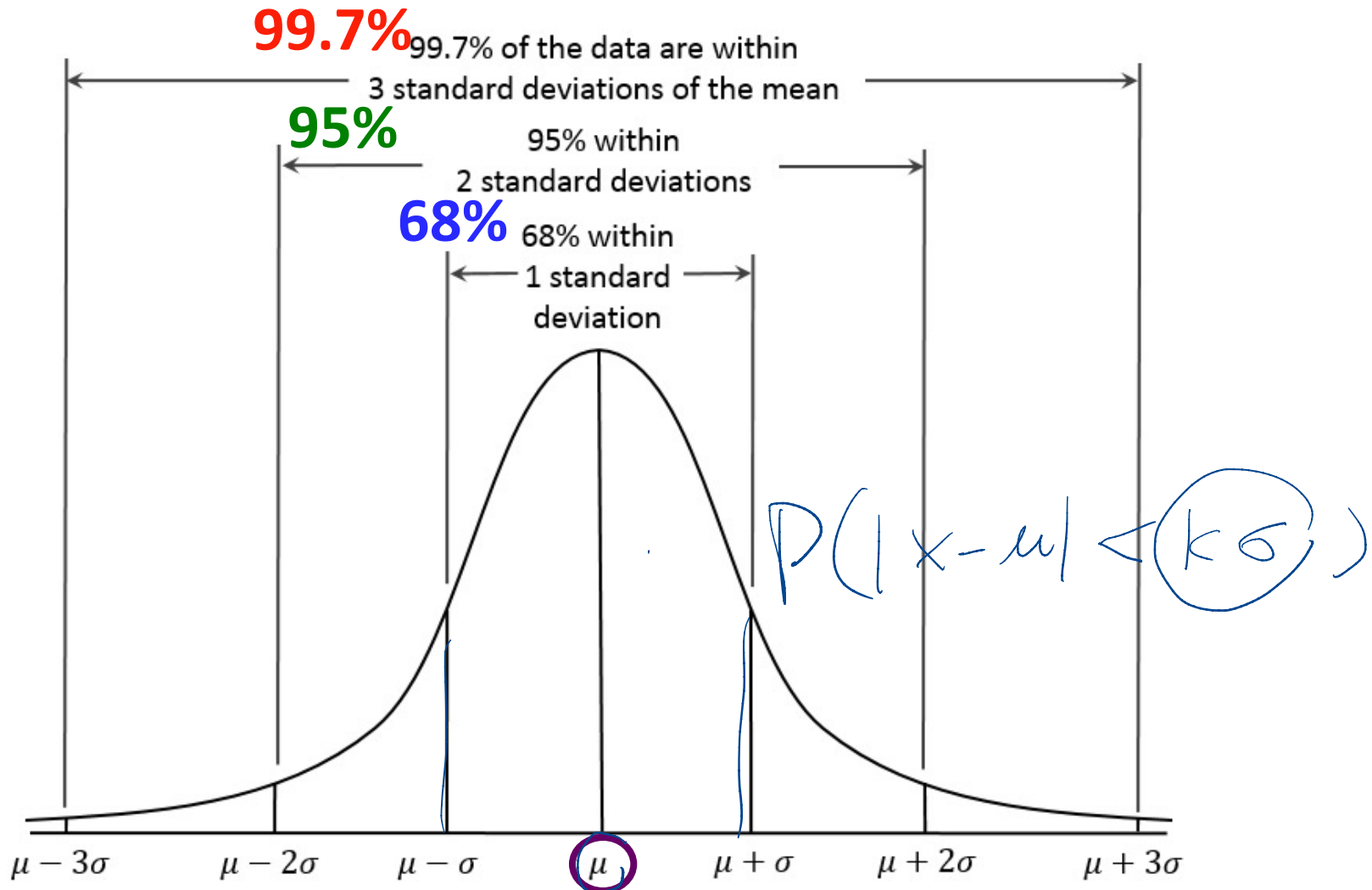
C. 1

50% quantile

for $p(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}$

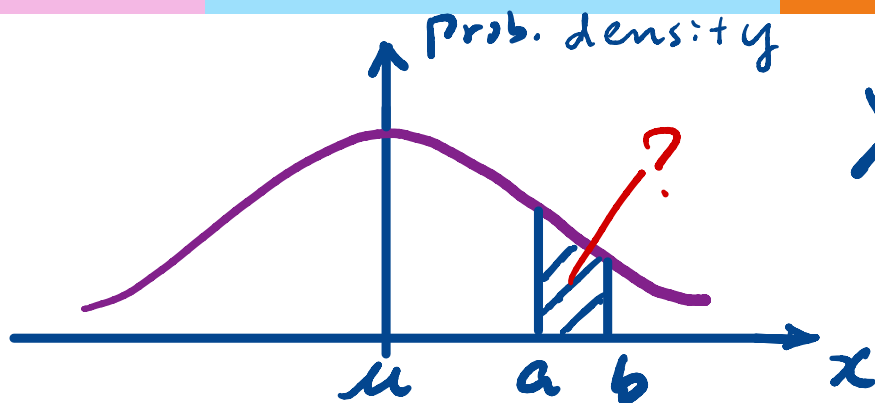


Spread of normal (Gaussian) distributed data



Credit:
wikipedia

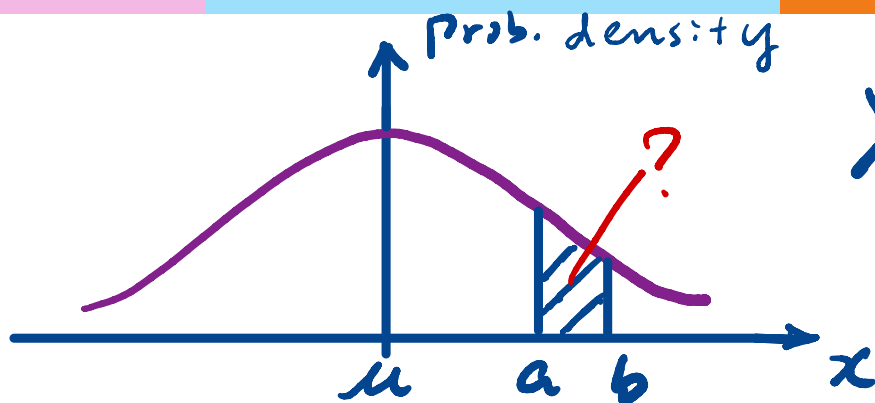
What is this probability?



$$X \sim N(\mu, \sigma^2)$$

$$\int_a^b p(x) dx = F(b) - F(a)$$

What is this probability?



$$X \sim N(\mu, \sigma^2)$$

$$\begin{aligned} P(a < X < b) &= \int_a^b \frac{e^{-\left(\frac{x-\mu}{\sigma}\right)^2/2}}{\sqrt{2\pi} \sigma} dx \\ &= ? \end{aligned}$$

No analytical solution!

Standard normal distribution

- ✱ If we standardize the normal distribution (by subtracting μ and dividing by σ), we get a random variable that has standard normal distribution.
- ✱ A continuous random variable X is **standard normal** if

$$p(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)$$

$$\mu = 0$$

$$\sigma = 1$$

Standard normal distribution

- ✱ If we standardize the normal distribution (by subtracting μ and dividing by σ), we get a random variable that has standard normal distribution.
- ✱ A continuous random variable X is **standard normal** if

$$p(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)$$

$$E[X] = 0 \quad \& \quad \text{var}[X] = 1$$

Derivation of standard normal distribution

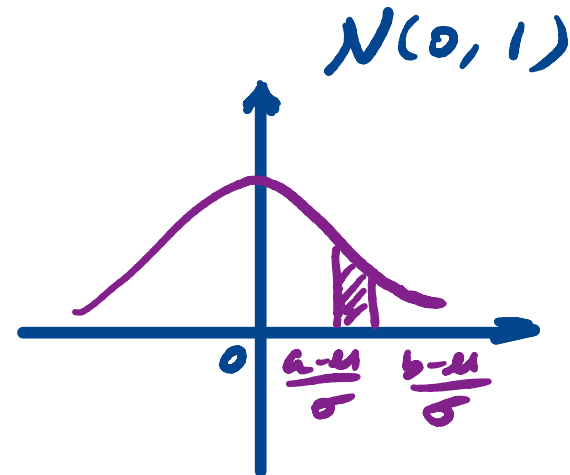
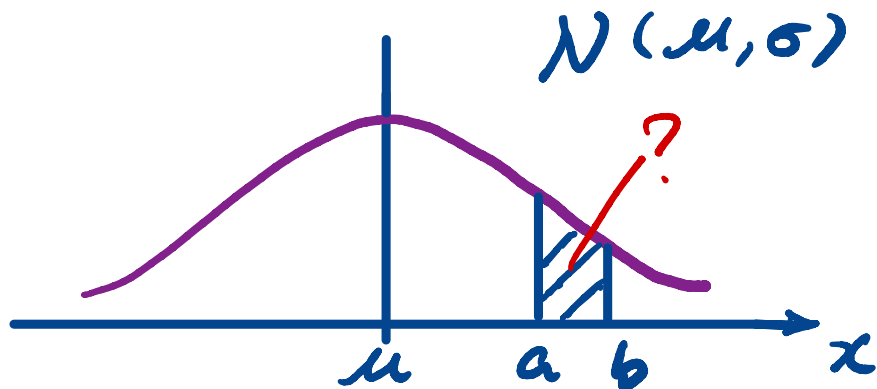
$$\begin{aligned}
 & \int_{-\infty}^{+\infty} p(x) dx \\
 = & \int_{-\infty}^{+\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx \\
 = & \int_{-\infty}^{+\infty} \frac{1}{\cancel{\sigma}\sqrt{2\pi}} \exp\left(-\frac{\hat{x}^2}{2}\right) \cancel{\sigma} d\hat{x} \\
 = & \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{\hat{x}^2}{2}\right) d\hat{x} \\
 = & \int_{-\infty}^{+\infty} p(\hat{x}) d\hat{x}
 \end{aligned}$$

$\hat{x} = \frac{x - \mu}{\sigma}$
 $\frac{x - \mu}{\sigma}$
 $\hat{\mu} = 0$
 $\hat{\sigma} = 1$

Call this standard and omit using a **hat**

$$p(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)$$

What is this probability?



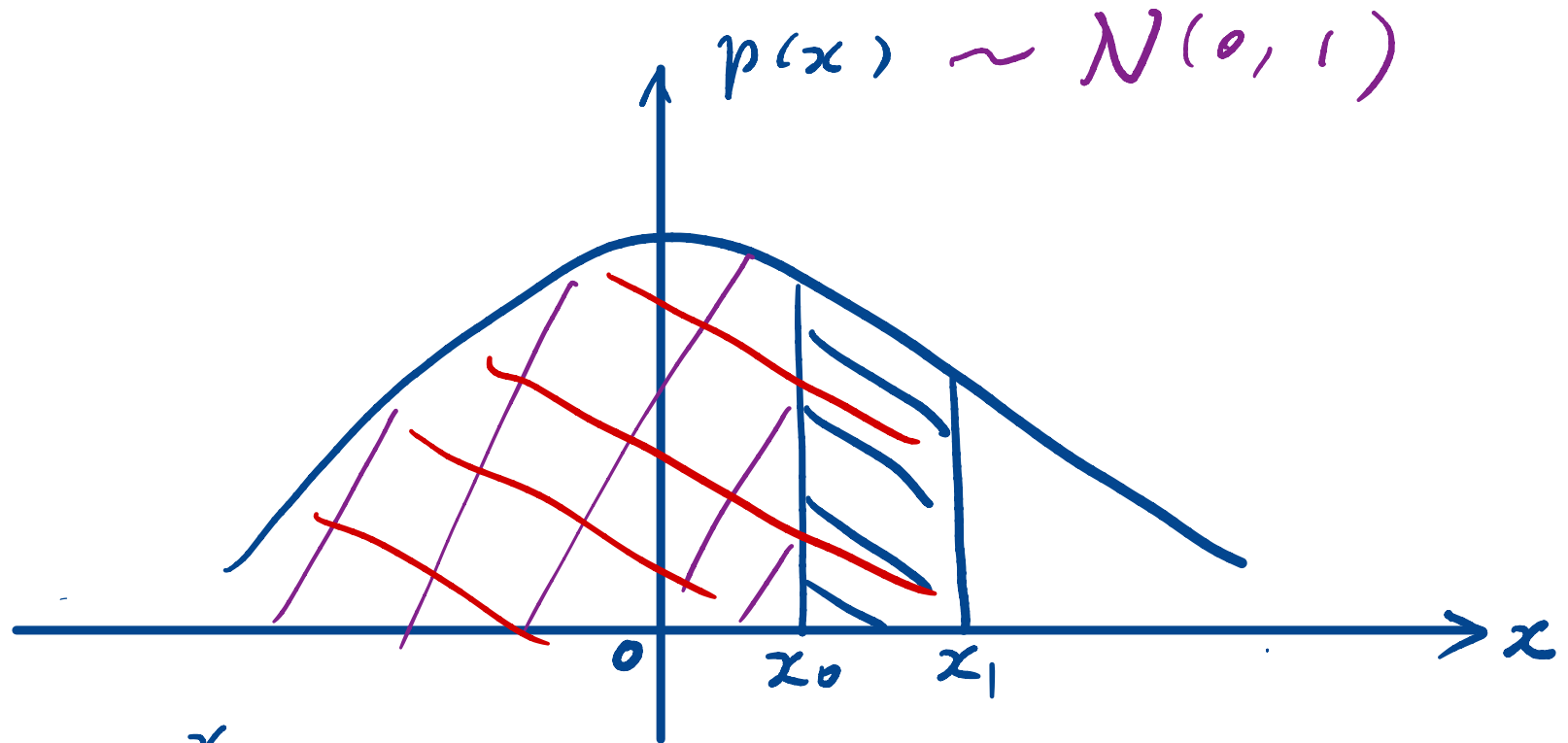
$$p(a < X < b) = \int_a^b \frac{e^{-\left(\frac{x-\mu}{\sigma}\right)^2/2}}{\sqrt{2\pi}\sigma} dx$$

$\hat{x} = \frac{x-\mu}{\sigma}$
 $d\hat{x} = \frac{dx}{\sigma}$

$$= \int_{\frac{a-\mu}{\sigma}}^{\frac{b-\mu}{\sigma}} \frac{e^{-\hat{x}^2/2}}{\sqrt{2\pi}} d\hat{x}$$
$$= F_{N(0,1)}\left(\frac{b-\mu}{\sigma}\right) - F_{N(0,1)}\left(\frac{a-\mu}{\sigma}\right)$$

No analytical sol.!

Calculate Prob. of Standard Normal



$$\int_{x_0}^{x_1} p(x) dx = \text{CDF}(x_1) - \text{CDF}(x_0)$$

$\downarrow$

$$P(X < x_1) - P(X < x_0)$$

Another way to check the spread of normal distributed data

- ✱ Fraction of **normal** data within **1** standard deviation from the mean.

$$\frac{1}{\sqrt{2\pi}} \int_{\frac{\mu-\sigma}{\sigma}}^{\frac{\mu+\sigma}{\sigma}} \exp\left(-\frac{x^2}{2}\right) dx \simeq 0.68$$

Handwritten notes: $\int_{\frac{\mu-\sigma}{\sigma}}^{\frac{\mu+\sigma}{\sigma}} \frac{e^{-x^2/2}}{\sqrt{2\pi}} dx$

- ✱ Fraction of **normal** data within **k** standard deviations from the mean.

$$\frac{1}{\sqrt{2\pi}} \int_{-k}^k \exp\left(-\frac{x^2}{2}\right) dx$$

Handwritten notes: $= \underset{\downarrow N(0,1)}{\text{CDF}(k)} - \underset{\downarrow N(0,1)}{\text{CDF}(-k)}$

Using the standard normal's table to calculate for a normal distribution's probability

✱ If $X \sim N(\mu=3, \sigma^2=16)$ (normal distribution)

$N(0, 1)$

$$\hat{X} = \frac{X - \mu}{\sigma} \quad P(X \leq 5) = ?$$

$$F(x) = CDF = \int_{-\infty}^x p(x) dx = P(X \leq x)$$

$$= \frac{X - 3}{4}$$

$$X = 3 + 4\hat{X}$$

$$3 + 4\hat{X} \leq 5$$

$$\hat{X} \leq \frac{2}{4}$$

$$= 0.5$$

	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015

Q.

✱ If $X \sim N(\mu=3, \sigma^2=16)$ (normal distribution)

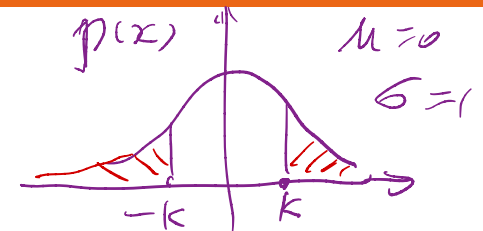
$P(X \leq 5) = ?$ A . 0.5199 B. 0.5987 **C. 0.6915**

	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
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1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015

Q. Is the table with only positive x values enough?

A. Yes

B. No.



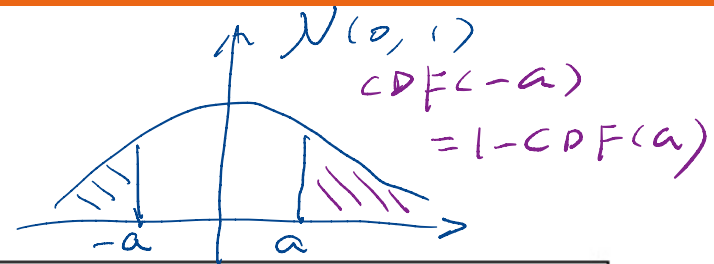
	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
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1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319

$N(0, 1)$
 $CDF(-k)$
 $= 1 - CDF(k)$

Q. Is the table with only positive x values enough?

A. Yes

B. No.



	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
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Using the standard normal's table to calculate for a normal distribution's probability

✱ If $X \sim N(\mu=3, \sigma^2=16)$ (normal distribution)

~~$P(X \leq 5) = ?$~~

$P(X^2 < 4) = ?$

$N(0, 1)$

$F(x) = \text{CDF} = \int_{-\infty}^x p(x) dx = P(X \leq x)$
 $N(0, 1)$

	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
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1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015

$\hat{X} = \frac{X - \mu}{\sigma}$

$= \frac{X - 3}{4}$

$X = 3 + 4\hat{X}$

$|\hat{X}| < 2$

$|3 + 4\hat{X}| < 2$

$$|3 + 4\hat{X}| < 2$$

$$-2 < 3 + 4\hat{X} < 2$$

$$-\frac{5}{4} < \hat{X} < -\frac{1}{4}$$

$$\begin{aligned} P(|3 + 4\hat{X}| < 2) &= P(\hat{X} < -\frac{1}{4}) - P(\hat{X} < -\frac{5}{4}) \\ &= \underset{N(0,1)}{CDF}(-\frac{1}{4}) - \underset{N(0,1)}{CDF}(-\frac{5}{4}) \\ &= \left(1 - CDF(\frac{1}{4})\right) - \left(1 - CDF(\frac{5}{4})\right) \\ &= CDF(\frac{5}{4}) - CDF(\frac{1}{4}) \\ &= 0.8944 - 0.5987 = 0.2957 \end{aligned}$$

Central limit theorem (CLT)

- ✱ The distribution of the average of N independent identical (IID) random variables tends toward a **normal** distribution as $N \rightarrow \infty$
- ✱ Even when the component random variables are not exactly IID, the result is approximately true and very useful in practice

$$\hat{z}_N = \left(\frac{\sum x_i}{N} \right)_{\text{standardized}} \quad F(\hat{z}_N) \rightarrow F(N(0,1))$$

read Degroot Pg 236-240
* 368-369

Central limit theorem (CLT)

- ✱ CLT helps explain the prevalence of normal distributions in nature
- ✱ A binomial random variable tends toward a normal distribution when ***N*** is large due to the fact it is the sum of IID Bernoulli random variables

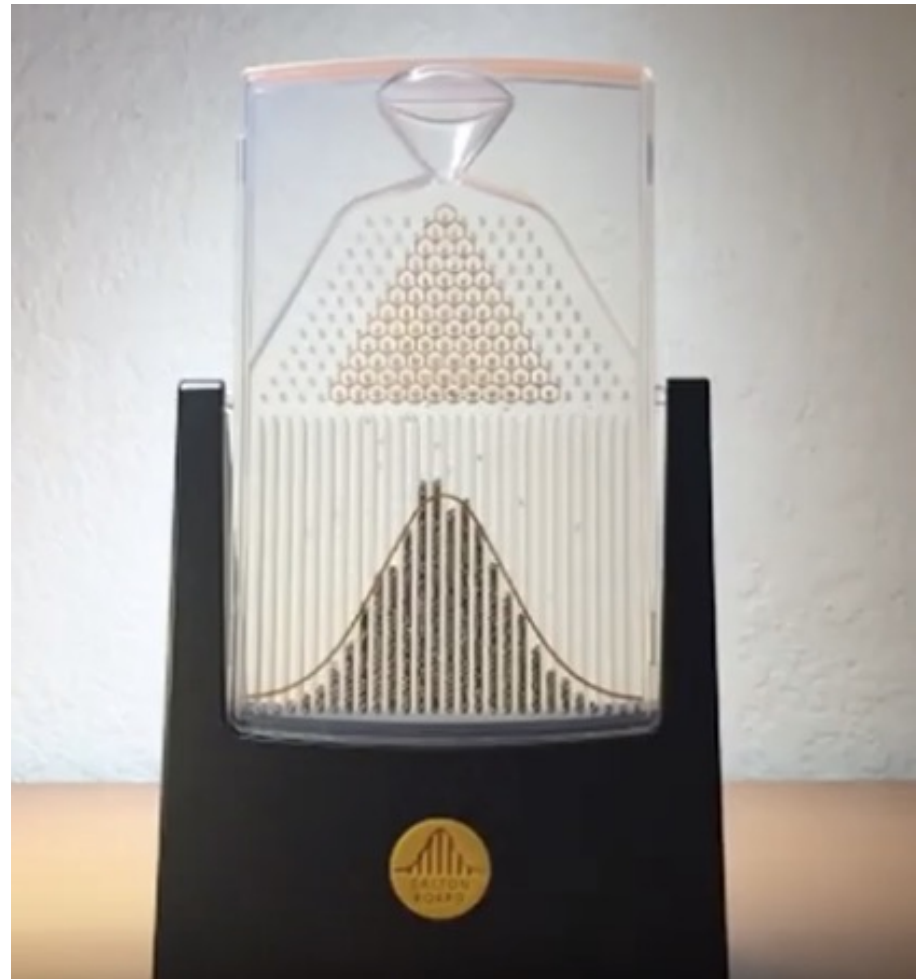
$$X_{\text{Bino}} = \sum_{i=1}^N X_{\text{Bei}}$$
$$\frac{X_{\text{Bino}}}{N} = \frac{\sum X_{\text{Bei}}}{N}$$

$$X_{\text{Bei}} = \begin{cases} 1 & \text{head } p(\text{head}) = p \\ 0 & \text{ow} \end{cases}$$

$$\sum_0^K P(X=k) \rightarrow \int_0^K p(x) dx$$

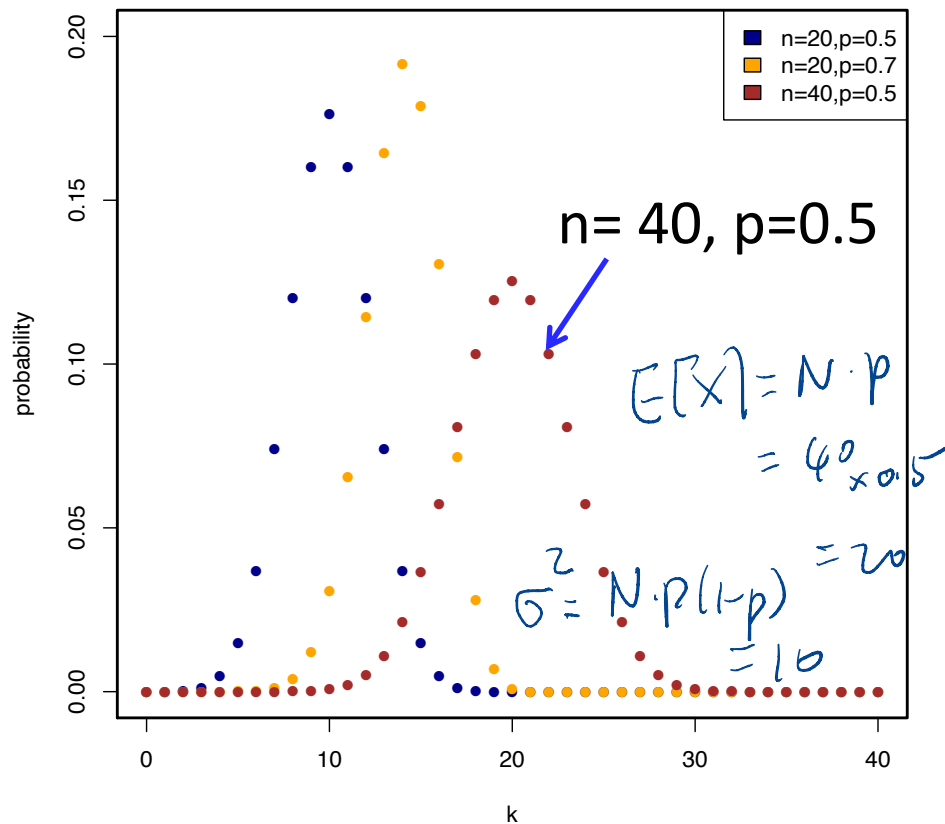
The Binomial distributed beads of the Galton Board

The **Binomial** distribution looks very similar to **Normal** when N is large

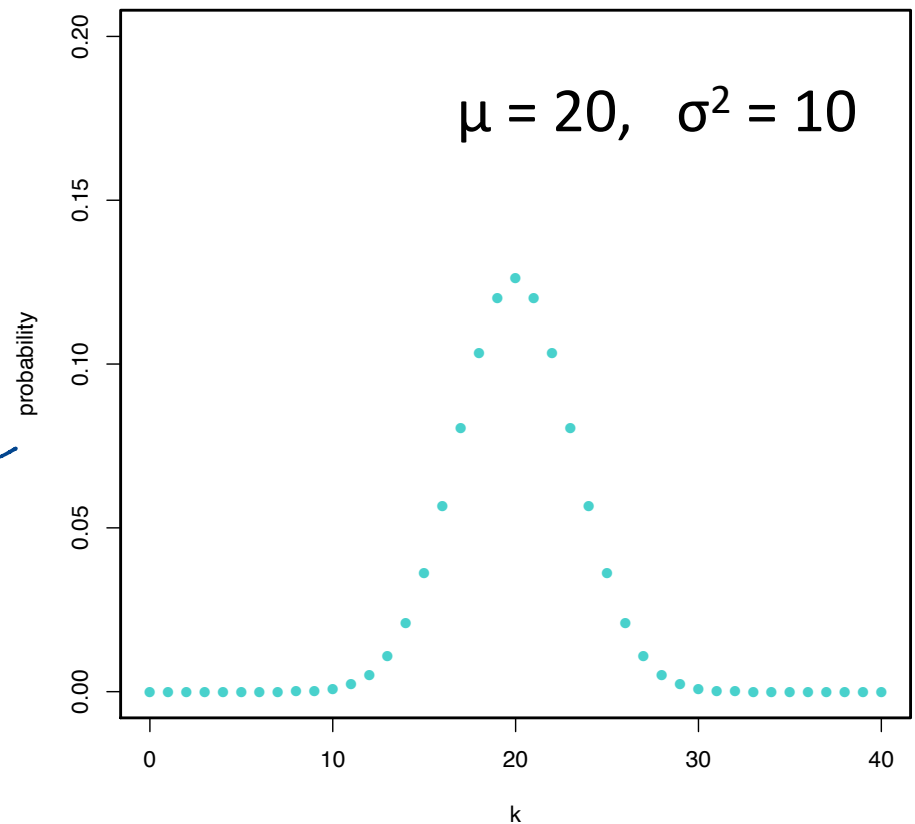


Binomial approximation with Normal

Binomial distribution



Approximation with Normal



Binomial approximation with Normal

- ✱ Let k be the number of heads appeared in 40 tosses of fair coin
- ✱ The goal is to estimate the following with normal

$$P(10 \leq k \leq 25) = \sum_{k=10}^{25} \binom{40}{k} 0.5^k 0.5^{40-k}$$

$$= \sum_{k=10}^{25} \binom{40}{k} 0.5^{40} \simeq 0.96$$

$$E[k] = np = 40 \cdot 0.5 = 20 \qquad \begin{aligned} std[k] &= \sqrt{np(1-p)} \\ &= \sqrt{40 \cdot 0.5 \cdot 0.5} = \sqrt{10} \end{aligned}$$

Binomial approximation with Normal

- ✱ Use the same mean and standard deviation of the original binomial distribution.

$$\mu = 20 \qquad \sigma = \sqrt{10} \simeq 3.16$$

- ✱ Then standardize the normal to do the calculation

$$P(10 \leq k \leq 25) \simeq \frac{1}{\sigma\sqrt{2\pi}} \int_{10}^{25} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{\frac{10-20}{3.16}}^{\frac{25-20}{3.16}} \exp\left(-\frac{x^2}{2}\right) dx$$

$$\simeq 0.94$$

$$= F_{N(0,1)}\left(\frac{25-20}{3.16}\right) - F_{N(0,1)}\left(\frac{10-20}{3.16}\right)$$

$$P(10 \leq X \leq 25)$$

Assignments this week

- ✱ Week 6 quiz open at 4:30pm
- ✱ HW5 due
- ✱ Prepare for Midterm1:
 - ✱ Practice exams
 - ✱ Read through instructions on Canvas

Additional References

- ✱ Charles M. Grinstead and J. Laurie Snell
"Introduction to Probability"
- ✱ Morris H. Degroot and Mark J. Schervish
"Probability and Statistics"

See you next time

*See
You!*

